Speaker Script

Bootstrap Inference in the Presence of Bias

Cavaliere, Gonçalves, Nielsen & Zanelli (JASA, 2024)

Jordi Torres — TSE Econometrics Reading Group

This document is a slide-by-slide companion to `presentation_bootstrapbias.tex`*. For each slide it contains: (i) a tight script, (ii) the notation introduced, (iii) the transition into the next slide, and (iv) anticipated aggressive questions with prepared answers. A target of ∼45 minutes implies roughly 1–1.5 minutes per slide.*

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1. Master cheat-sheet of notation

Define each symbol *once* below and use it throughout. After the relevant slide introduces it, do *not* re-define.

Population / data

$\theta \in \mathbb{R}$	scalar parameter of interest.
D_n	original sample of size n.
D_n^*	bootstrap sample drawn from a bootstrap DGP *conditional* on D_n.

Statistics

$\hat{\theta}_n$	estimator of θ based on D_n.
$\hat{\theta}_n^*$	bootstrap version of $\hat{\theta}_n$, computed from D_n^*.
$g(n)$	convergence rate ($n^{1/2}$, $(nh)^{1/2}$, etc.).
T_n	$:= g(n)(\hat{\theta}_n - \theta)$, the centred and rescaled statistic.
T_n^*	$:= g(n)(\hat{\theta}_n^* - \hat{\theta}_n)$, the bootstrap version (note centring at $\hat{\theta}_n$, not θ).

Bias terms (the heart of the paper)

B_n	deterministic asymptotic bias sequence: $B_n \to B$.
B	limiting bias of T_n (constant; *may not be consistently estimable*).
$\hat{B}_n$	*implicit* bootstrap bias — D_n-measurable, defined so that $T_n^* - \hat{B}_n$ has the same asymptotic noise as $T_n - B_n$. Crucially *not* an estimator the researcher chooses; it is whatever bias the bootstrap DGP produces.
$\hat{B}_n^*$	second-level analogue used in the double bootstrap.

Limit objects

ξ_1 limit of $T_n - B_n$, centred at zero, cdf G continuous.

ξ_2 limit of $\hat{B}_n - B_n$, centred at zero. *Nonzero ξ_2 is the failure.*

G cdf of ξ_1.

F cdf of $\xi_1 - \xi_2$ (continuous by assumption).

H $:= F \circ G^{-1}$, asymptotic cdf of $\hat{p}_n$. Continuous, free of B.

Bootstrap p-values

$\hat{p}_n$ $:= P^*(T_n^* \leq T_n)$, standard bootstrap p-value.

$\hat{p}_n^*$ second-level p-value $P^{**}(T_n^{**} \leq T_n^*)$ for the double bootstrap.

$\hat{H}_n$ estimator of H (plug-in or double bootstrap).

$\tilde{p}_n$ $:= \hat{H}_n(\hat{p}_n)$, the prepivoted (modified) p-value.

Modes of convergence

$\xrightarrow{p}$ convergence in probability under P.

$\xrightarrow{d}$ convergence in distribution under P.

P^* $:= P(\cdot \mid D_n)$, bootstrap probability measure (random in D_n).

$\xrightarrow{p}_p$ (written $\xrightarrow{p^*}_p$)
$Y_n^* \xrightarrow{p}_p 0$ means $P^*(|Y_n^*| > \varepsilon) \xrightarrow{p} 0$ for every $\varepsilon > 0$. (Convergence to 0 in bootstrap probability *in probability under P*.)

$\xrightarrow{d^*}_p$ $Y_n^* \xrightarrow{d^*}_p \xi$ means $P^*(Y_n^* \leq u) \xrightarrow{p} P(\xi \leq u)$ at every continuity point.

Read aloud only when first introduced. After the bootstrap-statistic slide, just say "in bootstrap probability" or "conditionally on the data."

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2. Part 1 — Introduction (slides 1–8)

Slide 1. *Title slide.*
Say: Today I am presenting Cavaliere, Gonçalves, Nielsen & Zanelli, JASA 2024. The paper is about how to do valid bootstrap inference when the estimator has an asymptotic bias the bootstrap cannot replicate. Before going into details, let me place the paper in our reading group.
Transition: Move to the roadmap.

Slide 2. *Roadmap.*
Say: Five blocks: a short *introduction* that fixes notation and states the problem; three *motivating examples*; the *general theory* — Theorem 3.1 and prepivoting — which is the core of the talk; the *examples revisited* where I show how the fix is implemented; and a short *wrap-up.*
Transition: Begin with the framing slide.

Slide 3. *Placing the paper in the reading group.*
Say: In the reading group so far we have seen two roles for the bootstrap. First, computing vari-

ances and full sampling distributions when analytical formulas are unavailable. Second, asymptotic refinements — Beran 1987 and Hall 1992 — where the bootstrap can be more accurate than the first-order normal approximation.

Today's paper is different. It asks: what if the estimator has an asymptotic bias B that the bootstrap simply cannot replicate? Then the standard bootstrap is invalid, even asymptotically. Cavaliere et al. show that valid inference can be restored *without estimating the bias* by a transformation of the bootstrap p-value, the prepivoting idea of Beran 1987, 1988.

Why this slide: This is the only narrative slide in the deck. From here on we go technical.

Transition: Now I make the setup precise.

Anticipated questions: Q: Why is bias different from variance? Cant we just bootstrap the bias?

A: Variance is a property of the sampling distribution that the bootstrap mimics by construction. Bias is the gap between the population parameter and the limit of the estimator; the bootstrap replicates the law of T_n^* *around* $\hat{\theta}_n$, not the law of T_n around θ, so bias does not transfer.

Q: Is this related to the failure of the bootstrap under the null in unit-root or moment-condition problems?

A: Conceptually yes — in all those cases the bootstrap measure is random in the limit. The unifying language here is condition (1.3) ahead.

Slide 4. *Setup — the original statistic (1.1).*

Say: Let $\theta \in \mathbb{R}$ be a scalar parameter, $\hat{\theta}_n$ an estimator with rate $g(n) \to \infty$. Define $T_n := g(n)(\hat{\theta}_n - \theta)$. Equation (1.1) says $T_n \xrightarrow{d} B + \xi_1$, with ξ_1 centred at zero and continuous cdf G.

The point: B is an *asymptotic bias*. In every example we will see, B is *not* consistently estimable. The typical setting is $g(n) = n^{1/2}$ and ξ_1 Gaussian, but the paper allows non-Gaussian ξ_1.

Why does classical inference fail? Because using quantiles of ξ_1 to build confidence intervals or tests *ignores* B, so coverage is wrong.

Why not just bias-correct analytically? Either B cannot be estimated (model averaging, ridge with local-to-zero parameter), or it can be estimated but only poorly (nonparametric regression: $\beta''(x)$ requires a second bandwidth). And as the box notes, the bootstrap also *cannot* solve the bias problem on its own when B is not estimable.

Notation introduced: $\theta, \hat{\theta}_n, T_n, B_n, B, \xi_1, G, g(n)$.

Transition: Now, what does the bootstrap give us?

Anticipated questions: Q: What is "not consistently estimable" precisely? Lots of biases are slow but estimable.

A: Take model averaging: bias is $Q \cdot c$ where $\delta = cn^{-1/2}$. The parameter c is unidentified — $\hat{\delta}_n = O_p(n^{-1/2})$ converges in *distribution*, not in probability, when scaled by $n^{1/2}$. So no estimator can be consistent for c. Same logic for ridge.

Q: Doesnt Hall (1992) handle this with Edgeworth refinements?

A: No — those refinements assume the bootstrap is first-order correct and then chase higher-order terms. Here it fails at first order.

Slide 5. *Setup — the bootstrap statistic (1.2).*

Say: $T_n^* := g(n)(\hat{\theta}_n^* - \hat{\theta}_n)$ — centred at $\hat{\theta}_n$ because, in the bootstrap world, $\hat{\theta}_n$ plays the role of the "true" parameter (the only feasible choice; θ is unknown). Equation (1.2): $T_n^* - \hat{B}_n \xrightarrow{d^*}_p \xi_1$ — the bootstrap reproduces the noise ξ_1 correctly.

Critically, $\hat{B}_n$ is *implicit*: it is whatever asymptotic bias the bootstrap DGP *produces*, not something the researcher chooses or estimates. You can think of $\hat{B}_n$ as "the bootstrap's own implicit estimator of B — even though no one wrote it down."

If $\hat{B}_n - B = o_p(1)$ — i.e. the implicit estimator is consistent — the bootstrap is valid in the usual sense. The whole question of the paper is: what happens when no such consistent implicit estimator can exist?

The notation block on the right defines two modes of bootstrap convergence I will use repeatedly.

Notation introduced: $T_n^*, \hat{B}_n, D_n, D_n^*, P^*, \xrightarrow{d^*}_p, \xrightarrow{p}_p$.

Transition: Now I state the failure mode.

Anticipated questions: **Q:** What does "implicit" mean? Where is $\hat{B}_n$ defined operationally?
A: Operationally $\hat{B}_n$ is whatever sequence makes (1.2) hold — it is a property of the bootstrap DGP, not a choice. In the model averaging FRB example we will see $\hat{B}_n = Q_n\, n^{1/2}\hat{\delta}_n$.

Q: Why centre T_n^* at $\hat{\theta}_n$ rather than θ?
A: Because θ is unknown; centring at $\hat{\theta}_n$ is the only feasible choice. This is also why the bootstrap distribution is conditional on D_n.

Slide 6. *The failure mode (1.3).*

Say: Condition (1.3): $\hat{B}_n - B \xrightarrow{d} \xi_2 \not\equiv 0$, jointly with (1.1). This is the failure mode — and the key idea worth dwelling on.

Why does this happen? $\hat{B}_n$ is built from the data, and in the problems we care about, the data noise sits at the *same rate* as the bias B. So no estimator of B from this data can be consistent — it inherits an irreducible $O_p(1)$ noise of the same order as the thing it is trying to estimate. The bootstrap's implicit $\hat{B}_n$ is no exception.

Two noises, not one. ξ_1 is the sampling noise in T_n — the bootstrap reproduces this. ξ_2 is the bootstrap's failure to pin down the bias — this is an *extra* source of randomness that the bootstrap cannot subtract out, because it *is* the bootstrap's own error.

Consequences. The bootstrap distribution of T_n^* conditional on D_n is itself *random in the limit* — it sits at a random shift $\hat{B}_n$, and that randomness ξ_2 never washes out. So T_n^* and T_n have *different* asymptotic laws (one centred at $\hat{B}_n$, the other at B_n, differing by ξ_2). Hence $\hat{p}_n := P^*(T_n^* \leq T_n)$ is *not* asymptotically uniform, and a nominal 5% test can reject at, say, 15%.

Counterfactual to keep in mind: if $\hat{B}_n - B_n = o_p(1)$ (so $\xi_2 = 0$), then T_n^* and T_n would have the same asymptotic law and $\hat{p}_n$ would be uniform — standard story. (1.3) is exactly the obstruction.

The goal of the paper: restore valid inference without trying to estimate B — by working with the bootstrap p-value $\hat{p}_n$ directly.

Notation introduced: ξ_2, $\hat{p}_n$.

Transition: The fix is on the next slide.

Anticipated questions: **Q:** Random in the limit — is this related to Cavaliere–Georgiev (2020) or to random measures?
A: Yes, exactly. CG 2020 showed randomness of the bootstrap measure does not by itself prevent validity. Here it does, because the asymptotic bias is also random.

Q: Could we just use a bigger bootstrap, m-out-of-n, to fix this?
A: Often m-out-of-n *kills* the bias in the bootstrap world, giving $\hat{B}_n \xrightarrow{p} 0$ and $\xi_2 = -B$ — a constant. By Remark 3.1 ahead, that case is *worse*, not better: the distortion now depends on B, so prepivoting no longer works either.

Slide 7. *A way out — prepivoting.*

Say: Beran's idea, 1987–88. Even though $\hat{p}_n$ is not uniform, its asymptotic cdf H is going to be *continuous* and, more importantly, *free of the unknown bias B*. If so, the probability integral transform gives $H(\hat{p}_n) \xrightarrow{d} U[0,1]$. Replace $\hat{p}_n$ by $\tilde{p}_n := \hat{H}_n(\hat{p}_n)$, with $\hat{H}_n$ uniformly consistent for H.

Two implementations: *plug-in* when H depends on a finite-dimensional parameter, and *double bootstrap* when it does not. Beran proposed this for refinements; here it rescues a broken bootstrap.

Notation introduced: $H, \hat{H}_n, \tilde{p}_n$.

Transition: Three motivating examples.

Anticipated questions: **Q:** Why should H be free of B in general?
A: This is the content of Theorem 3.1, which I will prove. The reason is that both numerator and denominator in $H = F \circ G^{-1}$ are cdfs of *centred* limits ξ_1 and $\xi_1 - \xi_2$, and B is the centring that has been removed by definition.

Q: Is prepivoting a refinement or a fix?
A: Beran used it for refinements — improving second-order accuracy. Here it fixes a first-order failure. Mechanically the operation is the same.

3. Part 2 — Motivating examples (slides 9–14)

Slide 9. *Three motivating examples (lead-in).*

Say: For each example I will check four things in order: the model and parameter; whether (1.1) holds; whether (1.2) holds; and whether (1.3) holds. The third one — (1.3) — is the key to understanding why the bootstrap fails.

Transition: Example 1, model averaging.

Slide 10. *Example 1 — Ridge regression, setup.*

Say: Linear regression with p regressors. Test a linear restriction $g'\theta = r$. Ridge estimator with shrinkage c_n added to the diagonal of S_{xx}. The local-to-zero device is $\theta = \delta n^{-1/2}$ and $c_n/n \to c_0 > 0$. As before, this is the regime where shrinkage bias matters.

Anticipated questions: **Q:** $c_n \sim n$ is a strong tuning. Is that standard?
A: It is the regime studied by Fu and Knight (2000) and Hjort and Claeskens (2003). The point is to keep shrinkage at the same order as noise. Other rates lead to either no shrinkage in the limit or to $\theta = 0$ being imposed asymptotically.

Slide 11. *Example 1 — Why (1.1)–(1.3) hold.*

Say: (1.1) — bias survives in the limit. From $\tilde{S}_{xx}^{-1} S_{xx} - I = -(c_n/n)\tilde{S}_{xx}^{-1}$, T_n splits into a shrinkage piece (limit $B \propto c_0\,\delta$) and a Gaussian noise piece (limit $\xi_1 \sim N(0, v^2)$). The bias depends on the local-to-zero δ, which is *not consistently estimable* because $n^{1/2}\hat{\theta}_n = \delta + O_p(1)$ converges in *distribution*, not in probability.

(1.2) — bootstrap captures the noise. Pairs bootstrap, *centred at OLS*, not at ridge. Reason: the bootstrap world's "true parameter" must satisfy the score condition $E^*[x_t^* \varepsilon_t^*] = 0$; OLS does, ridge does not. Same recentring trick as in the GMM bootstrap (Horowitz, Hall–Horowitz).

(1.3) — and here is the key point. The implicit bootstrap bias is $\hat{B}_n = -(c_n/n) g' \tilde{S}_{xx}^{-1}\, n^{1/2}\hat{\theta}_n$. Subtracting B_n:

$$\hat{B}_n - B_n = -\tfrac{c_n}{n} g' \tilde{S}_{xx}^{-1}\, n^{1/2}(\hat{\theta}_n - \theta).$$

$c_n/n \to c_0 > 0$ and $n^{1/2}(\hat{\theta}_n - \theta) = O_p(1)$ *not* $o_p(1)$ — OLS is only $n^{1/2}$-consistent, so its noise does not vanish at the relevant scale. Multiplying $O_p(1)$ noise by a non-degenerate matrix gives $O_p(1)$. Hence $\xi_2 \neq 0$.

Tie to the general story: the bootstrap is implicitly trying to estimate B via $\hat{B}_n$, and $\hat{B}_n$ inherits the OLS noise — which is the *same order* as the bias itself. That is exactly why the bias is "random in the limit" in this example.

Transition: Example 2, nonparametric.

Anticipated questions: **Q:** Why pairs and not residual bootstrap?
A: Residual bootstrap requires homoskedasticity and a correctly specified mean function. Pairs is robust. The recentring at OLS is the technical price.

Q: Wouldnt this also fail under wild bootstrap?
A: Yes. The failure is again about $\hat{\theta}_n - \theta$ being only $n^{1/2}$-consistent, not $o(n^{-1/2})$.

Slide 12. *Example 2 — Nonparametric regression, setup.*

Say: Nonparametric regression with smooth $\beta(\cdot)$, fixed design. Nadaraya–Watson with MSE-optimal bandwidth $h \sim n^{-1/5}$.

The MSE-optimal h balances squared bias against variance. At this rate, *the bias does not vanish faster than the noise*; in fact it shows up in the limit. That is the source of (1.1).

Anticipated questions: **Q:** Why not undersmooth?
A: Undersmoothing kills bias asymptotically but at the cost of larger variance and lower power against local alternatives. CCF 2018 document this trade-off.

Slide 13. *Example 2 — Why (1.1)–(1.3) hold.*
Say: (1.1) — bias survives in the limit. Taylor-expand the kernel smoother: limit bias $B \propto \beta''(x)$, limit noise $\xi_1 \sim N(0, v^2)$. The $\beta''(x)$ is estimable in principle, but only with a second bandwidth — unreliable in finite samples (CCF 2018). *Effectively inestimable in practice.*

(1.2) — bootstrap captures the noise. Parametric bootstrap with $y_t^* = \hat{\beta}_h(x_t) + \varepsilon_t^*$, $\varepsilon_t^* \sim N(0, \hat{\sigma}_n^2)$.

(1.3) — here is the key point, with a *different mechanism but the same moral.* The bootstrap bias replaces $\beta(\cdot)$ with $\hat{\beta}_h(\cdot)$, so it depends on $\hat{\beta}_h''(x)$ instead of $\beta''(x)$. The estimation error is

$$\hat{\beta}_h''(x) - \beta''(x) = O_p((nh)^{-1/2}),$$

exactly the same order as the bias itself at the MSE-optimal bandwidth (that is what "MSE-optimal" means: bias and noise rates are balanced). Hence $\hat{B}_n - B_n$ does not vanish, and the bootstrap's implicit estimate of B inherits irreducible $O_p(1)$ noise.

Tie to the general story: same as ridge — $\hat{B}_n$ is built from data whose noise sits at the same rate as B. The mechanism is different ($O_p(1)$ OLS noise in ridge vs. MSE-optimal smoothing here), but the conclusion is identical: the bootstrap's bias is *random in the limit.*

Cross-example punchline: **whenever bias and noise are at the same rate, no estimator of B from the data can be consistent — the bootstrap inherits this and (1.3) bites.**

Transition: Recap, then theory.

Anticipated questions: **Q:** Couldnt one use a higher-order kernel to remove B?
A: Yes, in principle, but at higher variance and at the cost of negative weights and finite-sample instability. Standard MSE-optimal kernels are second order.

Slide 14. *Examples — recap.*
Say: Same diagnostic in both: rate, source of bias, the inestimable quantity that makes (1.3) bite, the bootstrap scheme. Different mechanisms ($O_p(1)$ nuisance vs. MSE-optimal smoothing) but same conclusion. Now we step back and develop a unified theory.

Transition: Section 3 of the paper.

4. Part 3 — General results (slides 15–24)

Slide 15. *Recap of notation.*
Say: Quick refresher of the symbols I will use throughout this section. The bottom block is the most important: $\xrightarrow{d^*}_p$ means convergence in distribution conditional on the data, in probability.

G, F, H will appear constantly: G is the limit cdf of $T_n - B_n$; F is the limit cdf of $T_n - \hat{B}_n$, which equals the cdf of $\xi_1 - \xi_2$; $H = F \circ G^{-1}$ will be the asymptotic cdf of $\hat{p}_n$.

Transition: The two assumptions.

Slide 16. *Assumption 1 — the original statistic.*
Say: Assumption 1 is just (1.1) restated, plus regularity on G: continuous and strictly increasing on its support.

Continuous, because Polya's theorem upgrades pointwise to uniform convergence of cdfs — I will use that in the proof of Theorem 3.1.

Strictly increasing, because we need G^{-1} well-defined to apply the probability integral transform.

Importantly, G does *not* need to be Gaussian. Stable distributions are allowed.

Anticipated questions: **Q:** What if G is discontinuous — discrete data?
A: Then the framework needs randomisation as in Romano (1990). Out of scope here.

Slide 17. *Assumption 2 — the bootstrap side.*

Say: Two parts. (i) is (1.2): the bootstrap captures the noise correctly. (ii) is the joint convergence of $(T_n - B_n, \hat{B}_n - B_n)$ to (ξ_1, ξ_2), with F continuous.

Two things to flag. First, $\hat{B}_n$ is again *not* a chosen estimator. It is whatever the bootstrap DGP produces. Second, F only needs to be continuous — not strictly increasing.

Anticipated questions: Q: Joint convergence is strong — when does it fail?

A: It can fail under non-stationary or near-unit-root settings; see Cavaliere– Georgiev for examples. In all five examples in the paper, joint convergence is straightforward because everything is linear in the same score.

Slide 18. *Benchmark: what if* $\hat{B}_n$ were *consistent?*

Say: Useful warm-up: assume $\hat{B}_n - B_n = o_p(1)$, so $\xi_2 = 0$ and $F = G$. Then $\hat{p}_n = P^*(T_n^* - \hat{B}_n \leq T_n - \hat{B}_n) \to G(T_n - \hat{B}_n) + o_p(1) \xrightarrow{d} G(\xi_1) \sim U[0, 1]$.

So when $\xi_2 = 0$ the bootstrap is uniform: the standard story. The *non-trivial* case is $\xi_2 \neq 0$, where $F \neq G$ and the limit is $G(F^{-1}(U)) \neq U$. That is exactly Theorem 3.1.

Transition: State and prove Theorem 3.1.

Slide 19. *Theorem 3.1.*

Say: Under Assumptions 1–2, $\hat{p}_n \xrightarrow{d} G(F^{-1}(U))$. Three implications. $\hat{p}_n$ is not uniform unless $G = F$. The asymptotic cdf is $H = F \circ G^{-1}$. And H does not depend on B — the bias has been subtracted away in both F and G. *This is the algebraic miracle that makes the fix feasible.*

Anticipated questions: Q: Why doesnt B appear in H?

A: Both F and G are cdfs of *centred* variables (ξ_1 and $\xi_1 - \xi_2$ respectively). Centring removes B by construction. The information about B is in the location of T_n itself, not in H.

Slide 20. *Theorem 3.1 — proof.*

Say: Four steps. *Step 1.* Subtract $\hat{B}_n$ from both sides inside P^*; $\hat{B}_n$ is constant under P^* since it is D_n-measurable. By Assumption 2(i) and Polya, $P^*(T_n^* - \hat{B}_n \leq u) \to G(u)$ uniformly. So $\hat{p}_n = G(T_n - \hat{B}_n) + o_p(1)$.

Step 2. Decompose $T_n - \hat{B}_n = (T_n - B_n) - (\hat{B}_n - B_n)$. By Assumption 2(ii) and CMT, this converges jointly to $\xi_1 - \xi_2$.

Step 3. G is continuous, so $G(T_n - \hat{B}_n) \xrightarrow{d} G(\xi_1 - \xi_2)$.

Step 4. $\xi_1 - \xi_2$ has continuous cdf F, so by the probability integral transform $\xi_1 - \xi_2$ has the same distribution as $F^{-1}(U)$. Plugging in gives $\hat{p}_n \xrightarrow{d} G(F^{-1}(U))$. Done.

Anticipated questions: Q: Why is Polya needed?

A: To upgrade pointwise convergence of $P^*(T_n^* - \hat{B}_n \leq u)$ to uniform. We then evaluate at the random argument $T_n - \hat{B}_n$ and use Slutsky. Without continuity of G, the substitution does not go through.

Q: Where do you actually use joint convergence of (1.1) and (1.3)?

A: In Step 2: the CMT applied to the difference requires joint distributional convergence, not just marginal.

Slide 21. *Three remarks.*

Say: Remark 3.1 is the most important. Some bootstrap schemes set $\hat{B}_n \xrightarrow{p} 0$ — they structurally remove the bias in the bootstrap world. Then $\xi_2 = -B$, a constant. The distortion of $\hat{p}_n$ now *depends on* B. So prepivoting cannot fix it: H would depend on the unknown B, and you cannot estimate it. *Lesson: prefer bootstrap schemes that try to capture the bias, even noisily.*

Remarks 3.2 and 3.3 are corollaries: confidence intervals are similarly invalid, and the bootstrap conditional distribution is random in the limit.

Transition: Now use Theorem 3.1 to fix the problem — Corollary 3.1.

Anticipated questions: **Q:** Concretely, which schemes give $\hat{B}_n = 0$?
A: The m-out-of-n bootstrap for heavy tails (paper's Example 4); the cross-sectional pairs bootstrap for dynamic panels (Example 5). Both are discussed in the supplement.

Slide 22. *Corollary 3.1.*

Say: $H(\hat{p}_n) \xrightarrow{d} U[0,1]$. Proof: by Theorem 3.1, $\hat{p}_n \xrightarrow{d} V$ where $V = G(F^{-1}(U))$. Apply $H = F \circ G^{-1}$ to V: the inner Gs cancel, leaving $F \circ F^{-1}(U) = U$. By CMT, $H(\hat{p}_n) \xrightarrow{d} U$.

The reading is that applying H *unwarps* the distorted $\hat{p}_n$ back to uniform — exactly what we wanted. Of course, H is unknown, so we need $\hat{H}_n$.

Transition: Two ways to estimate H: plug-in, then double bootstrap.

Slide 23. *Approach A — Plug-in (Corollary 3.2).*

Say: Assume $H = H_\gamma$ for some finite-dimensional γ, with $\hat{\gamma}_n \xrightarrow{p} \gamma$. Corollary 3.2: if $H_\gamma(u)$ is jointly continuous in (γ, u), then $\tilde{p}_n := H_{\hat{\gamma}_n}(\hat{p}_n) \xrightarrow{d} U[0,1]$.

The proof is two pieces. First, by joint continuity in γ, $\sup_u |H_{\hat{\gamma}_n}(u) - H_\gamma(u)| \xrightarrow{p} 0$, so $\tilde{p}_n - H(\hat{p}_n) \xrightarrow{p} 0$. Second, $H(\hat{p}_n) \xrightarrow{d} U[0,1]$ by Corollary 3.1. Slutsky combines them.

The plug-in approach is cheap — one bootstrap plus one formula — but requires deriving H_γ in closed form. In the Gaussian case, which covers our examples, H takes a simple shape; that is the next slide.

Transition: Gaussian case — the variance-ratio correction.

Anticipated questions: **Q:** If $\hat{\gamma}_n$ is only $n^{1/2}$-consistent, do we need any rate of convergence?
A: No, only consistency in probability. The proof goes through Slutsky; no rate is required.

Slide 24. *Gaussian plug-in — the correction m.*

Say: Suppose $\xi_1 \sim N(0, v^2)$ and (ξ_1, ξ_2) are jointly normal. Write $\xi_1 - \xi_2 \sim N(0, v_d^2)$. Then $G(u) = \Phi(u/v)$, $F(u) = \Phi(u/v_d)$, and

$$H(u) \;=\; \Phi(m^{-1}\Phi^{-1}(u)), \qquad m \;=\; v_d/v.$$

Reading. Without (1.3), $T_n - \hat{B}_n \xrightarrow{d} N(0, v^2)$ — the bootstrap spread is v^2. Under (1.3), the actual spread is $v_d^2 \neq v^2$. The correction $\tilde{p}_n = \Phi(m^{-1}\Phi^{-1}(\hat{p}_n))$ rescales the p-value as if the spread had always been v_d^2. *B drops out completely* — only standard deviations appear.

In our two examples: ridge has m^2 depending on Σ_{xx} and c_0, but *not* on the unknown local-to-zero δ — so m is estimable. Nonparametric is even better: σ^2 cancels and m depends *only on the kernel* K — a known constant, no estimation at all.

Anticipated questions: **Q:** Why does σ^2 cancel only in nonparametric?
A: In the nonparametric case both v and v_d scale linearly with σ^2, so the ratio is dimensionless in σ. In ridge the correlation between ξ_1 and ξ_2 is different and σ^2 does not cancel out.

Q: Is m always less than 1, or always greater?
A: Generally $m \geq 1$ in these examples: $\xi_1 - \xi_2$ has more spread than ξ_1 because of the extra randomness from ξ_2. So the correction shrinks $\Phi^{-1}(\hat{p}_n)$ toward zero, making the test more conservative.

Q: Why is this a "ratio of standard deviations" rather than something with B?
A: Because both G and F are cdfs of *centred* variables; B has been subtracted away by definition. So H depends only on the variance structure of (ξ_1, ξ_2), not on the bias.

Slide 25. *Approach B — Double bootstrap.*

Say: If you cannot find H_γ analytically, use a double bootstrap. From each first-level sample D_n^*, draw a second-level D_n^{**}. Compute the second-level p-value $\hat{p}_n^*$. Estimate H by the bootstrap cdf of $\hat{p}_n^*$. $\tilde{p}_n$ is then $P^*(\hat{p}_n^* \leq \hat{p}_n)$ — the fraction of second-level p-values below the original.

Theorem 3.2 says this works under Assumptions 1–3. Assumption 3 just says the double bootstrap replicates both the noise (at level 2) and the bias gap.

Transition: Corollary 3.3 gives a much cheaper alternative.

Anticipated questions: **Q:** Computational cost?
A: B^2 where B is the bootstrap size — typically prohibitive. That is why Corollary 3.3 matters.

Slide 26. *Corollary 3.3 — the practical shortcut.*
Say: The double bootstrap modified p-value is *asymptotically equivalent* to a single-level bootstrap applied to the bias-corrected statistic $T_n - \hat{B}_n$. So if $\hat{B}_n$ is available in closed form (model averaging, ridge), no double bootstrap is needed: just bootstrap the bias-corrected statistic.

Note: $\hat{B}_n$ does *not* need to be a consistent estimator of B_n. We are not estimating bias — we are using whatever the bootstrap provides as the centring device.

Transition: Examples revisited.

Anticipated questions: **Q:** Doesnt this look like just bias-correcting?
A: No. We are not bias-correcting in the sense of plugging in an estimator of B. We are using $\hat{B}_n$ as a centring; $\hat{B}_n$ is biased for B_n in general (that is the whole point). The validity comes from Theorem 3.2, not from $\hat{B}_n \to B$.

5. Part 4 — Examples revisited (slides 27–28)

Slide 27. *Ridge — implementing the fix.*
Say: Same recipe. The variance ratio simplifies to a ratio involving Σ_{xx} and the regularised version $\tilde{\Sigma}_{xx}$. Sanity check: when $c_0 = 0$ (no shrinkage), $m = 1$ and there is no correction — ridge collapses to OLS, which is unbiased.

Slide 28. *Nonparametric — implementing the fix.*
Say: The cleanest case: m depends *only on the kernel*, no estimation needed. Compute $\hat{\beta}_h(x)$ at the MSE-optimal bandwidth, run the parametric bootstrap, look up m from the kernel, report $\tilde{p}_n = \Phi(m^{-1}\Phi^{-1}(\hat{p}_n))$.

Compare to existing approaches. Undersmoothing kills the bias but at the cost of power. Robust bias correction (CCF 2018) needs a second bandwidth. This paper: keep the MSE-optimal h, one-line correction, no estimation of $\beta''(x)$. Cleaner and finite-sample friendly.

Transition: Wrap-up.

Anticipated questions: **Q:** Why does the paper not include simulations?
A: They are in the supplement.

6. Part 5 — Wrap-up

Slide 29. *Summary.*
Say: Four bullets.

Problem. Asymptotic bias B that the bootstrap cannot replicate makes $\hat{p}_n$ non-uniform.

Diagnosis. The bootstrap bias $\hat{B}_n$ misses B by a non-vanishing random ξ_2. Source: an inestimable quantity at the $n^{-1/2}$ boundary.

Fix. The asymptotic cdf $H = F \circ G^{-1}$ of $\hat{p}_n$ is continuous and free of B. So $\tilde{p}_n = \hat{H}_n(\hat{p}_n) \xrightarrow{d} U[0,1]$.

Two routes. Plug-in (Gaussian: variance ratio $m = v_d/v$) or double bootstrap. Corollary 3.3 reduces the latter to a single bootstrap of $T_n - \hat{B}_n$.

The bigger picture for the reading group: a third role for the bootstrap. Not variance, not refinement, but *rescuing inference* when the first-order bootstrap is invalid — without ever estimating the bias.

Transition: Questions.

7. Catch-all questions to be ready for

Q: Why not just do uniform inference using subsampling (Politis–Romano)?
A: Subsampling is consistent under weaker conditions, but it can be very imprecise in finite samples and choosing the subsample size b is delicate. Prepivoting requires stronger assumptions but, when applicable, gives a clean closed-form correction (especially in the Gaussian case).

Q: Is this the same as Andrews–Guggenberger size-correction?
A: Both target asymptotic non-uniformity, but the mechanisms differ. Andrews–Guggenberger correct for non-uniform convergence over the parameter space; Cavaliere et al. correct for a random bootstrap distribution in the limit at a single parameter value.

Q: What happens under heteroskedasticity / dependence?
A: Use a bootstrap that handles it (wild, block) and re-derive Assumption 2. The framework does not change; only the verification of (1.2) and (1.3).

Q: Could H depend on a nuisance parameter that is itself not consistently estimable?
A: Yes, in principle. The plug-in approach then fails, but the double bootstrap can still work — it estimates H nonparametrically. This is exactly the selling point of double bootstrap.

Q: Why prepivoting and not simply bootstrap-of-bootstrap on the studentised statistic?
A: Studentisation only fixes scale, not bias. The non-uniformity here is location-driven (random $\hat{B}_n - B$). Studentisation does not address that.

Q: If $\hat{B}_n$ is implicit, how do we even compute the correction in practice?
A: For Gaussian limits we never compute $\hat{B}_n$ directly; we estimate the *variances* v^2 and v_d^2. For Corollary 3.3 we do need a closed form for $\hat{B}_n$, which the paper provides for ridge and model averaging.

Q: How robust is the prepivoted test to misspecification?
A: Same robustness as the underlying bootstrap. Prepivoting is a transformation; it does not strengthen distributional assumptions but inherits them.

8. Modes of Convergence Used in the Paper

Standard convergence

Definition (Convergence in probability)**.** $Y_n \xrightarrow{p} \xi$ means: for every $\varepsilon > 0$,

$$P(|Y_n - \xi| > \varepsilon) \to 0 \quad \text{as } n \to \infty.$$

The random variable Y_n gets arbitrarily close to ξ with probability approaching 1. When $\xi = c$ is a constant, this is the usual consistency statement (e.g. $\hat{\sigma}_n^2 \xrightarrow{p} \sigma^2$).

Definition (Convergence in distribution)**.** $Y_n \xrightarrow{d} \xi$ means: for every continuity point u of the cdf of ξ,

$$P(Y_n \le u) \to P(\xi \le u).$$

This is weaker than convergence in probability: Y_n and ξ need not be close in any single realisation; only their *distributions* converge. The CLT is the canonical example: $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} N(0,1)$.

Bootstrap convergence

Definition (Bootstrap weak convergence in probability: $\xrightarrow{d^*}_p$)**.** $Y_n^* \xrightarrow{d^*}_p \xi$ means: for every continuity point u of the cdf $G(u) = P(\xi \le u)$,

$$P^*(Y_n^* \le u) - G(u) \xrightarrow{p} 0.$$

Here $P^*(\cdot) = P(\cdot \mid D_n)$ is the **conditional** probability given the original data D_n. The quantity $P^*(Y_n^* \le u)$ is itself a random variable (it depends on which D_n was observed). The statement says: this random cdf converges *in probability* (over realisations of D_n) to the deterministic cdf G.

Why "in probability"? Different realisations of D_n give different bootstrap distributions. For most datasets the bootstrap cdf will be close to G, but for some pathological datasets it might not be. The "in probability" qualifier says the set of bad datasets has vanishing probability.

Equivalently: $Y_n^* = o_{p^*}(1)$ means $P^*(|Y_n^*| > \varepsilon) \xrightarrow{p} 0$ for all $\varepsilon > 0$.

Definition (Double bootstrap convergence: $\xrightarrow{d^{**}}_{p^*}$, in probability)**.** $Y_n^{**} \xrightarrow{d^{**}}_{p^*} \xi$, in probability, means: for every continuity point u of G,

$$P^{**}(Y_n^{**} \le u) - G(u) \xrightarrow{p^*}_{p} 0.$$

Now $P^{**}(\cdot) = P(\cdot \mid D_n^*, D_n)$ is conditional on both the first-level bootstrap data D_n^* and the original data D_n. The convergence $\xrightarrow{p^*}_p 0$ means: the P^*-probability that $|P^{**}(Y_n^{**} \le u) - G(u)|$ exceeds ε converges to zero in P-probability.

In plain language: for most original datasets D_n, and for most first-level bootstrap datasets D_n^* drawn from D_n, the second-level bootstrap cdf is close to G.

How they relate in the paper

Statement	What's random	Conditioned on
$T_n - B_n \xrightarrow{d} \xi_1$	T_n (through D_n)	nothing
$T_n^* - \hat{B}_n \xrightarrow{d^*}_p \xi_1$	T_n^* (through bootstrap variates)	D_n
$T_n^{**} - \hat{B}_n^* \xrightarrow{d^{**}}_{p^*} \xi_1$	T_n^{**} (through level-2 variates)	D_n^* and D_n

Each level adds one layer of conditioning. The "in probability" qualifiers handle the fact that the conditioning set is itself random.

9. Two Key Theorems Used in the Proofs

Polya's theorem

Remark (Polya's theorem)**.** If $F_n(u) \to F(u)$ pointwise and F is **continuous**, then the convergence is **uniform**:

$$\sup_{u \in \mathbb{R}} |F_n(u) - F(u)| \to 0.$$

Why the paper needs this. In the proof of Theorem 3.1, we have $P^*(T_n^* - \hat{B}_n \le u) \to G(u)$ pointwise (from Assumption 2(i)). We then evaluate this at the *random* point $u = T_n - \hat{B}_n$. Pointwise convergence alone does not let you plug in a random argument — the error could be large exactly where the random point lands. Uniform convergence (via Polya, using continuity of G) guarantees the error is small *everywhere simultaneously*, so plugging in a random point is safe:

$$|\hat{p}_n - G(T_n - \hat{B}_n)| \le \sup_u |P^*(T_n^* - \hat{B}_n \le u) - G(u)| \xrightarrow{p} 0.$$

This is why Assumption 1 requires G continuous. Without continuity, Polya does not apply, uniform convergence fails, and the proof breaks at Step 1.

The probability integral transform (PIT)

Remark (PIT)**.** If X has continuous cdf F, then $F(X) \sim U[0,1]$.

Proof in one line: $P(F(X) \leq u) = P(X \leq F^{-1}(u)) = F(F^{-1}(u)) = u$.

The paper uses the PIT twice:

(1) In the **benchmark** (consistency case): $G(\xi_1) \sim U[0,1]$ because ξ_1 has cdf G.

(2) In **Corollary 3.1**: $H(\hat{p}_n) \xrightarrow{d} U[0,1]$ because $\hat{p}_n$ has asymptotic cdf H.

This is why G must be strictly increasing (for use 1: ensures G^{-1} exists cleanly) and H **must be continuous** (for use 2: ensures the transform works for the p-value).

10. Ridge Regression Example — Full Walkthrough

Why ridge? Where is it used?

Ridge regression (and penalised estimation more broadly: LASSO, elastic net) is the standard tool when you have many regressors with limited individual explanatory power. You shrink coefficients toward zero to reduce variance at the cost of some bias. This is now ubiquitous in applied econometrics — high-dimensional controls in causal inference, machine learning for prediction, regularised IV, etc.

The model

Linear regression $y_t = \theta' x_t + \varepsilon_t$, $t = 1, \ldots, n$, where x_t is a $p \times 1$ non-stochastic vector and $\varepsilon_t \sim \text{iid}(0, \sigma^2)$.

Interest is in testing $H_0 : g'\theta = r$ for a known vector g (e.g. $g = e_j$ tests the j-th coefficient).

The ridge estimator

The ridge estimator minimises the penalised least squares criterion:

$$\frac{1}{n}\|y - X\theta\|^2 + \frac{c_n}{n}\|\theta\|^2$$

giving the closed-form solution:

$$\tilde{\theta}_n = \left(S_{xx} + \frac{c_n}{n} I_p\right)^{-1} S_{xy} =: \tilde{S}_{xx}^{-1} S_{xy}$$

where $S_{xx} = X'X/n$ and $S_{xy} = X'y/n$.

What does ridge do mechanically? OLS inverts S_{xx}; ridge inverts $S_{xx} + \frac{c_n}{n} I_p$, a *larger* matrix (in the positive-definite sense). Inverting a larger matrix gives smaller coefficients — ridge systematically pulls $\tilde{\theta}_n$ toward zero. When $\theta \neq 0$, this pull is bias.

The source of bias

The key algebraic identity:

$$\tilde{S}_{xx}^{-1} S_{xx} - I_p = -\frac{c_n}{n} \tilde{S}_{xx}^{-1}$$

This follows from $S_{xx} = \tilde{S}_{xx} - \frac{c_n}{n} I$, so $\tilde{S}_{xx}^{-1} S_{xx} = I - \frac{c_n}{n} \tilde{S}_{xx}^{-1}$.

Using $S_{xy} = S_{xx}\theta + X'\varepsilon/n$:

$$\tilde{\theta}_n - \theta = \underbrace{-\frac{c_n}{n} \tilde{S}_{xx}^{-1} \theta}_{\text{bias (shrinkage)}} + \underbrace{\tilde{S}_{xx}^{-1} \frac{X'\varepsilon}{n}}_{\text{noise}}$$

The bias is proportional to $\frac{c_n}{n}$ (penalty strength) times $\tilde{S}_{xx}^{-1}\theta$ (shrinkage direction). This is entirely mechanical — ridge pulls $\tilde{\theta}_n$ toward zero, and the amount of pull depends on how large θ is and how strong the penalty is.

The local-to-zero framework

Set $\theta = \delta n^{-1/2}$ and $c_n/n \to c_0 > 0$.

Why this parametrisation?

- If θ is fixed: the normalised bias $n^{1/2} \cdot \frac{c_n}{n}\tilde{S}_{xx}^{-1}\theta \to \infty$. Ridge is terrible. Not interesting.
- If $\theta = 0$: no bias. Ridge = OLS asymptotically. Not interesting either.
- If $\theta = \delta n^{-1/2}$: the normalised bias $\frac{c_n}{n}\tilde{S}_{xx}^{-1}\delta \to c_0\tilde{\Sigma}_{xx}^{-1}\delta$, a nonzero constant. *This is the boundary where bias and noise are the same asymptotic order* — the regime where shrinkage genuinely helps but also genuinely distorts inference.

The test statistic and (1.1)

Multiply by $n^{1/2}$ and apply g':

$$T_n := n^{1/2}(g'\tilde{\theta}_n - r) = \underbrace{-\frac{c_n}{n}g'\tilde{S}_{xx}^{-1}\delta}_{B_n \to B = -c_0 g'\tilde{\Sigma}_{xx}^{-1}\delta} + \underbrace{g'\tilde{S}_{xx}^{-1}\frac{X'\varepsilon}{n^{1/2}}}_{\xi_{1,n} \xrightarrow{d} N(0,v^2)}$$

where $v^2 = \sigma^2 g'\tilde{\Sigma}_{xx}^{-1}\Sigma_{xx}\tilde{\Sigma}_{xx}^{-1}g$.

This confirms (1.1): $T_n - B_n \xrightarrow{d} \xi_1 \sim N(0, v^2)$ with $B = -c_0 g'\tilde{\Sigma}_{xx}^{-1}\delta \neq 0$.

Why B cannot be consistently estimated

To estimate $B = -c_0 g'\tilde{\Sigma}_{xx}^{-1}\delta$, we can estimate c_0, $\tilde{\Sigma}_{xx}$, and g consistently. But δ requires:

$$n^{1/2}\hat{\theta}_n = n^{1/2}\theta + n^{1/2}(\hat{\theta}_n - \theta) = \delta + O_p(1).$$

The $O_p(1)$ noise never vanishes — we observe δ plus irreducible noise, and no amount of data can separate them.

The bootstrap and (1.2)

Pairs bootstrap: resample (y_t^*, x_t^*) iid from (y_t, x_t). Compute $\tilde{\theta}_n^* = \tilde{S}_{x^*x^*}^{-1}S_{x^*y^*}$ and

$$T_n^* = n^{1/2}g'(\tilde{\theta}_n^* - \hat{\theta}_n).$$

Why centre at $\hat{\theta}_n$ (OLS), not $\tilde{\theta}_n$ (ridge)? In the bootstrap world, the "true parameter" is whatever makes bootstrap errors uncorrelated with regressors. OLS satisfies $\sum x_t(y_t - x_t'\hat{\theta}_n) = 0$, so $E^*[x_t^*\varepsilon_t^*] = 0$. Ridge does not satisfy this first-order condition — centring at $\tilde{\theta}_n$ would create spurious correlation. Same principle as GMM bootstrap (Horowitz).

The implicit bootstrap bias is $\hat{B}_n = -\frac{c_n}{n}g'\tilde{S}_{xx}^{-1}n^{1/2}\hat{\theta}_n$ (using $\tilde{S}_{x^*x^*}^{-1} - \tilde{S}_{xx}^{-1} = o_{p^*}(1)$). Then:

$$T_n^* - \hat{B}_n \xrightarrow{d^*}_p N(0, v^2).$$

This confirms (1.2): the bootstrap captures the noise correctly.

The failure: (1.3)

$$\hat{B}_n - B_n = -\frac{c_n}{n} g' \tilde{S}_{xx}^{-1} \cdot n^{1/2}(\hat{\theta}_n - \theta) \xrightarrow{d} \xi_2 \sim N(0, v_{22}), \quad v_{22} > 0.$$

This holds because $n^{1/2}(\hat{\theta}_n - \theta) = O_p(1)$, not $o_p(1)$. The bootstrap bias $\hat{B}_n$ tracks B_n with irreducible noise of the same order as B_n itself.

Key structural insight: both $T_n - B_n$ and $\hat{B}_n - B_n$ are linear in the same score $n^{1/2} S_{x\varepsilon}$:

$$\begin{pmatrix} T_n - B_n \\ \hat{B}_n - B_n \end{pmatrix} = \begin{pmatrix} g' \tilde{S}_{xx}^{-1} \\ -c_n n^{-1} g' \tilde{S}_{xx}^{-1} S_{xx}^{-1} \end{pmatrix} n^{1/2} S_{x\varepsilon} \xrightarrow{d} N(0, V)$$

They are jointly normal and correlated because they share the same source of randomness.

The fix

Lemma 4.4 verifies Assumptions 1 and 2 with $(\xi_1, \xi_2) \sim N(0, V)$. The distortion parameter simplifies to:

$$m^2 = \frac{g' \Sigma_{xx}^{-1} g}{g' \tilde{\Sigma}_{xx}^{-1} \Sigma_{xx} \tilde{\Sigma}_{xx}^{-1} g}$$

using the algebraic identity $\Sigma_{xx} + 2c_0 I + c_0^2 \Sigma_{xx}^{-1} = \tilde{\Sigma}_{xx} \Sigma_{xx}^{-1} \tilde{\Sigma}_{xx}$.

Plug-in recipe:

1. Compute ridge $\tilde{\theta}_n$ and $T_n = n^{1/2}(g' \tilde{\theta}_n - r)$.
2. Run pairs bootstrap: resample (y_t^*, x_t^*), centre at OLS: $T_n^{*(b)} = n^{1/2} g' (\tilde{\theta}_n^{*(b)} - \hat{\theta}_n)$. Compute $\hat{p}_n = B^{-1} \sum_b \mathbf{1}\{T_n^{*(b)} \le T_n\}$.
3. Estimate m from sample moments: $\hat{m}_n^2 = (g' S_{xx}^{-1} g)/(g' \tilde{S}_{xx}^{-1} S_{xx} \tilde{S}_{xx}^{-1} g)$.
4. Report $\tilde{p}_n = \Phi(\hat{m}_n^{-1} \Phi^{-1}(\hat{p}_n))$.

Sanity checks:

- $c_0 = 0 \Rightarrow \tilde{\Sigma}_{xx} = \Sigma_{xx} \Rightarrow m = 1$: no correction needed (ridge = OLS).
- m depends on Σ_{xx} and c_0 (estimable), *not* on δ (inestimable). The unknown bias B drops out completely.
- $\hat{m}_n \xrightarrow{p} m$ because $S_{xx} \to \Sigma_{xx}$ and $\tilde{S}_{xx} \to \tilde{\Sigma}_{xx}$.

Lemma 4.5 verifies Assumption 3, so the double bootstrap is also valid.